FIND IT — what the question is asking, common errors, scope

WHAT IS THIS QUESTION ASKING

IF THE QUESTION SAYS OR SHOWS	IT IS	GO TO
a shell diagram, or asks for protons, neutrons, valence electrons	atomic structure	2 ATOMS
"electron configuration", or an ion with a charge	configuration, and how ions differ	2 ATOMS
"molecular shape", "bond angle", VSEPR, a formula like BrF_5	shape from electron pairs	2 ATOMS
"ionic or covalent", "polar", electronegativity	bonding and polarity	2 ATOMS
g/mol, "molar mass", "how many moles", "how many molecules"	the mole	3 MOLES
a gas with P, V, T, or "at STP"	gas laws	3 MOLES
mol/L, "concentration", "diluted to", a burette, a titration	solutions	3 MOLES
"equilibrium", K_c, K_p, "shifts", "Le Chatelier"	equilibrium	4 ACIDS
pH, pOH, $[H^+]$, acid, base, K_a, buffer	**the most common question in the paper**	4 ACIDS, the pH tree
"given that..." then two or three equations with ΔH	Hess's law	5 ENERGY
ΔH, ΔS, ΔG, "spontaneous", "exothermic"	thermodynamics	5 ENERGY
"rate", a collision picture, a Maxwell-Boltzmann curve, a catalyst	kinetics	6 ELECTRONS
"oxidation number/state", "oxidised", "half-equation"	redox	6 ELECTRONS
two half-cells, a salt bridge, E° values, anode/cathode	electrochemistry	6 ELECTRONS

OUTSIDE THE A105 SYLLABUS

TOPIC	NOT EXAMINED	WHAT IS USED INSTEAD
Moles and gases	**STP** and 22.4 L/mol **Gay-Lussac** the **combined gas law**	molar volume is **24 dm³** at 20°C. Two-state problems run **PV = nRT twice**
Composition	**percent composition** **empirical** and molecular formula	absent from A105 entirely
Concentration	**ppm** **% w/v**	the four units are % by mass, g/L, mol/kg, mol/L
Equilibrium	**ICE tables** **K_p** and partial pressures leaving pure solids out of K_c	every equilibrium one are given is homogeneous
Acids	**K_w** as a symbol $K_a \times K_b = K_w$ the "x is small" approximation pH of a salt	**pH + pOH = 14** is given directly. Weak acids are always supplied with a **% dissociation**
Energy	**calorimetry**, q = mcΔT	not examined
Cells	**cell notation** $Zn\|Zn^{2+}\|\|Cu^{2+}\|Cu$ the **Nernst** equation	everything is at standard conditions
Lesson 11	**no deck and no worksheet exists**	if it is examinable, the material is not in what one were given – ask

COMMON ERRORS, ALL SIX SIDES

WHERE	THE TRAP	THE FIX
gases	Celsius into a gas law	**K = °C + 273**, every time
gases	dm³ where m³ is needed	R = 8.314 wants **m³** and **Pa**
titration	the mole ratio upside down	the bigger **coefficient** gets the bigger mole count
acids	stopping at pOH	**pH = 14 − pOH**
acids	the strong-acid method on a weak acid	a ⇌ or a quoted K_a means weak
buffers	the ratio inverted	**salt over acid**. More salt → pH above pK_a
Hess	reversing without flipping the sign of ΔH	reverse flips the **sign**, scaling changes the **size**
Gibbs	kJ mixed with J	ΔS tables are in **J**. Divide by 1000
Gibbs	−T × (−ΔS) read as minus	two negatives make a **plus**
redox	the oxidising agent	it is the thing that gets **reduced**. The two words must be opposites

COMMON ERRORS, ALL SIX SIDES

WHERE	THE TRAP	THE FIX
cells	multiplying E° when one scale a half-equation	**never scale the volts**
electrolysis	time in minutes or hours	Q = I × t needs **seconds**
everywhere	answering to the wrong precision	the energy worksheet asks for **2 decimal places**. Show the working either way

READING THE STEM — WHAT EACH INSTRUCTION REQUIRES

HOW THE MARKS DIVIDE

A question ending **"explain your answer"** carries the decision for one mark and a separate reason for each remaining mark. A **4**-mark question therefore needs the answer plus **3** distinct reasons, written as separate sentences.

COMMAND WORDS

"State" and "identify" require the name only. "Explain" requires what happens **and** why it follows. "Show your working" means the arithmetic line itself carries a mark.

Precision: thermodynamics answers are given to **2 decimal places**. Elsewhere match the precision of the data supplied, and state the **unit** every time — several worksheets award a mark for it alone.

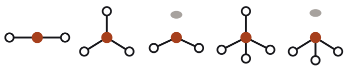
UNITS, PRECISION AND THE MARKS ATTACHED TO THEM

WHERE	CONVENTION	NOTE
gas calculations	V in **m³**, P in **Pa**, T in **K**	required by R = 8.314
solution calculations	V in **litres** for M = n ÷ V	but mL cancels in $M_1V_1 = M_2V_2$ and in titration
temperature	**K = °C + 273**	used throughout, not 273.15
enthalpy	tabulated data **kJ/mol**, calculated answers **kJ**	unless an enthalpy of formation is asked for
entropy	**$J\ mol^{-1}\ K^{-1}$**	**divide by 1000 before combining with an enthalpy**
equilibrium constant	$M^{\Delta n}$	Δn = 0 gives no units
rate constant	$M^{(1 - \text{overall order})}\ s^{-1}$	first order is s^{-1}
electrolysis	t in **seconds** for Q = It	convert back to hours if asked
thermodynamics answers	**2 decimal places**	elsewhere, match the precision of the data
every numerical answer	**state the unit**	several questions award a mark for the unit alone

ATOMS — structure, configuration, bonding, shapes

VSEPR — SHAPE FROM THE ELECTRON PAIRS

DOMAINS / LONE PAIRS	SHAPE	ANGLE, AND AN EXAMPLE
2 0 lone	**Linear**	180° CO_2
3 0 lone	**Trigonal planar**	120° BF_3
3 1 lone	**Bent or angular**	< 120° SO_2, O_3
4 0 lone	**Tetrahedral**	**109.5°** CH_4
4 1 lone	**Trigonal pyramid**	< 109.5°, about 107.5° NH_3
4 2 lone	**Bent or angular**	much < 109.5°, about 104.5° H_2O
5 0 lone	**Trigonal bipyramid**	90° **and** 120° PCl_5
5 1 lone	**Seesaw** (sawhorse)	< 90° and < 120° SF_4
5 2 lone	**T-shape**	< 90° ClF_3
5 3 lone	**Linear**	180° XeF_2
6 0 lone	**Octahedral**	90° SF_6
6 1 lone	**Square pyramid**	< 90° BrF_5
6 2 lone	**Square planar**	**exactly 90°** XeF_4
6 3 lone	**T-shape**	< 90°



Grey lobe = a lone pair. Each one squeezes the angle further, being the strongest.

GETTING TO THE SHAPE

SIGNATURE	METHOD	EXAMPLE
"what is the molecular shape of..."	**1.** draw the Lewis structure **2.** count **electron domains** on the central atom **3.** count how many of those are **lone pairs** **4.** read that cell of the table	CO_2: two double bonds → **2** domains → linear
what counts as one domain	**a double or triple bond is ONE domain**, same as a single	counting CO_2 as 4 gives tetrahedral, which is wrong
a quick route for AX_n with H, F, Cl, Br or I	lone pairs = (valence electrons of A − n) ÷ 2, then domains = n + lone pairs	BrF_5: (7−5)/2 = 1 lone, 6 domains → **square pyramid**
"electron geometry" vs "molecular shape"	geometry is the **0-lone-pair** name for that row. Shape is the actual cell	XeF_4 is **octahedral** geometry but **square planar** shape
"will the angles differ?"	**lone–lone > lone–bond > bond–bond** repulsion. More lone pairs, smaller angle	1 lone → about 107.5° 2 lone → about 104.5°. **Different**, and quote the numbers
a wedge-and-dash drawing	plain line in the plane, **bold wedge** forwards, **dashed** backwards	

ATOMS, IONS AND ISOTOPES

SIGNATURE	METHOD	EXAMPLE
a shell diagram, or a box with blanks	**1.** protons = atomic number, the **top** number in the box **2.** nuclear charge = **+Z**, write the plus sign **3.** neutrons = mass number − Z **4.** electrons = Z **for a neutral atom only** **5.** valence electrons = the outermost ring **6.** configuration fills **2, then 8, then 8**	boron: 5p, 6n, mass 11, **2,3**, charge 0
configuration of an **ion**	cation loses electrons, anion gains them. **Protons and mass do not change**	Mg^{2+}: still 12p and 12n, but **10 electrons → 2,8**
identify an atom from a diagram	count the **+** in the nucleus — that is the element. Then compare electrons to protons for the charge	more electrons than protons → anion
isotope abundances → relative atomic mass	RAM = Σ(isotope mass × its fraction)	Mg: 23.985(0.7899) + 24.986(0.1000) + 25.983(0.1101) = **24.31**
counts given, not percentages	**divide by the total first**	329+829+529 = 1687, so 329/1687 = 19.50%
abundances unknown, RAM given	let x be the lighter fraction, solve one equation	10.013x + 11.009(1−x) = 10.881 → x = **12.88%**
balance a **nuclear** equation	top numbers (mass) balance, bottom numbers (Z) balance, then read the element off Z. alpha is mass 4 / Z 2, beta is mass 0 / Z −1, positron is mass 0 / Z +1	U-238 → ? + alpha mass 238−4 = **234**, Z 92−2 = **90** → **Th-234**
beta or positron?	**the notes writes the positron with no plus sign**	beta is **−1** subscript, positron is **+1**. Misreading it changes the element

BONDING AND POLARITY

SIGNATURE	METHOD	EXAMPLE
"what type of bond"	find both electronegativities, subtract, read the band	**< 0.5** non-polar covalent **0.5–2.0** polar covalent **> 2.0** ionic
worked, all three bands	H–F, K–Cl, Cl–Cl	1.78 **polar covalent** 2.34 **ionic** 0 **non-polar**
"is the MOLECULE polar"	**1.** are the bonds polar? (ΔEN) **2.** what is the shape? **3.** do the bond dipoles **cancel**?	**polar bonds do not make a polar molecule.** CO_2 is linear, so they cancel → **non-polar**
symmetric AB_4 with identical B	dipoles cancel → **non-polar**	change one B, or add a lone pair, and it becomes polar
"which has the highest boiling point"	strongest intermolecular force wins: **ion-dipole > hydrogen bond > permanent dipole > temporary dipole**	

BONDING AND POLARITY

SIGNATURE	METHOD	EXAMPLE
does it hydrogen bond?	**the H must be bonded TO the F, O or N** — not merely present in the molecule	CH_3F contains F but every H is on carbon → **no** hydrogen bonding
electronegativity trend	increases across a period, decreases down a group	F is the most electronegative
atomic radius trend	decreases across a period, increases down a group	more shells → valence electrons further out and shielded

ACROSS: EN rises, radius falls
DOWN: EN falls, radius rises
F is the most electronegative element.

REFERENCE DATA NOT PRINTED ON THE PERIODIC TABLE

Ion charges are not tabulated, and one worksheet question needs it. The periodic table supplied shows atomic number, symbol, name and relative atomic mass — no charges. Group rule:

Gp 1 → 1+   Gp 2 → 2+   Gp 13 → 3+   Gp 15 → 3−
Gp 16 → 2−   Gp 17 → 1−   Gp 18 → 0

Then: metal + non-metal is **ionic**, two non-metals is **covalent**, and a covalent name tells one the count — **tri**chloride is 3, **tetra**fluoride is 4. So magnesium chloride is $MgCl_2$ and aluminium oxide is Al_2O_3.

Metallic bonding is not in these two lessons at all — the only three types the notes offers are non-polar covalent, polar covalent and ionic.

The notes disagrees with itself twice. The VSEPR image says the tetrahedral angle is 109° while another slide says **109.5°** — write 109.5°. And one slide calls relative atomic mass the "nucleon number", which it is not: relative atomic mass is a **weighted average** across isotopes, which is why boron is 10.81 and not a whole number.

WORKED — SHAPE, ANGLE AND POLARITY

Molecular shape of BrF_5

Determine the molecular shape and bond angle of BrF_5, and state whether the molecule is polar.

1. Valence electrons on the central atom: Br is in Group 17, so **7**
2. Five are used bonding to F, leaving 7 − 5 = 2 electrons = **1 lone pair**
3. Total electron domains = 5 bonds + 1 lone pair = **6**
4. Read the 6-domain, 1-lone-pair cell: **square pyramid**, angle **< 90°**
5. Polarity: Br–F bonds are polar (ΔEN = 3.98 − 2.96 = 1.02). The lone pair makes the shape **asymmetric**, so the dipoles do not cancel

Square pyramidal, bond angle < 90°, and the molecule is **polar**

MODEL WORDING — ATOMS AND BONDING

BOND ANGLE COMPARISON

The bond angle in **compound C** is **smaller**, because the central atom has **two** lone pairs rather than **one**. Lone pair to lone pair repulsion is greater than lone pair to bond pair repulsion, so the bonding pairs are pushed closer together.

BOND TYPE

The bond is **polar covalent**, because the difference in electronegativity is **1.78**, which lies between 0.5 and 2.0. The bonding pair is shared **unequally** between the two atoms.

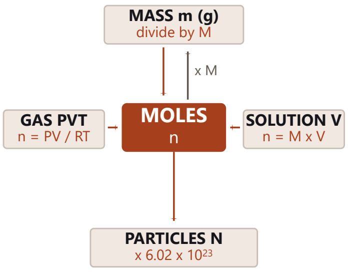
COMPARISON, GENERAL SHAPE

G:C requires more energy, because it is held by **three** hydrogen bonds while **A:T** has only **two**. More hydrogen bonds make the interaction stronger.

MOLES — molar mass, particles, the gas laws

UNITS AND CONSTANTS

QUANTITY	UNIT	RELATION
amount	**mol**	n = m ÷ M
mass, molar mass	**g, g/mol**	m = n × M
number of particles	atoms or molecules	N = n × **6.02 × 10^{23}**
temperature	**kelvin, always**	**K = °C + 273** (not 273.15)
gas volume, gas pressure	**m^3** and **Pa**	required by R = 8.314. dm^3 → m^3 is ÷1000, kPa → Pa is ×1000
solution volume	**L** (= dm^3)	M = n ÷ V
R, gas constant	**8.314** $m^3 \cdot Pa \cdot K^{-1} \cdot mol^{-1}$	the only value used; quoted in each question
molar volume	**24 dm^3/mol** at 20°C, 1 atm	recall only. Calculations run through PV = nRT
volume chain	1 dm^3 = 1 L = 1000 cm^3 = 1000 mL = **0.001 m^3**	appears in every part of a gas question



Every calculation goes through the middle.
Gas work in m3, Pa, K. Solutions in litres.

MOLES, MASS AND PARTICLES

SIGNATURE	METHOD	EXAMPLE
"molar mass of..."	**1.** count each atom, including subscripts **outside** brackets **2.** sum the relative atomic masses **3.** unit is **g/mol**	$CaCO_3$ = 40.08 + 12.01 + 3(16.00) = **100.09** $Al_2(SO_4)_3$: the 3 multiplies S **and** all four O → **342.17**
grams → moles	n = m ÷ M	102 g HBr ÷ 80.91 = **1.26 mol**
moles → grams	m = n × M	0.80 mol KBr × 119.00 = **95.20 g**
"how many molecules"	N = n × 6.02 × 10^{23}	2.00 mol F_2 → **1.20 × 10^{24}**
identify an unknown element	**1.** get n of the compound from the mole ratio **2.** M = mass ÷ n **3.** subtract the known atoms	M(XCl_2) = 110.98, less 2(35.45) = 40.08 → **Ca**

GASES

SIGNATURE	METHOD	EXAMPLE
moles change, T and P fixed	**Avogadro** $V_1/n_1 = V_2/n_2$	5 mol in 50 L → 6 mol = **60 L**. Only total moles matter
P and V, T fixed	**Boyle** $P_1V_1 = P_2V_2$	3 atm, 330 mL → 1 atm = **990 mL**
V and T, P fixed	**Charles** $V_1/T_1 = V_2/T_2$, T in K	25→70°C is 298→343 K. Increase 45/298 = **15.10%**
anything else	**PV = nRT**	n = PV/RT V = nRT/P T = PV/nR
both P and T change	**no combined gas law in A105.** Apply PV = nRT twice, or take the ratio	$V_2/V_1 = (P_1/P_2)(T_2/T_1)$
identify a gas from mass and PVT	**1.** n = PV/RT **2.** M = mass ÷ n **3.** match M to a formula	M = 0.71/0.0100 = 71 g/mol → **Cl_2**

WORKED — GAS LAW IN BASE UNITS

Pressure from PV = nRT

1.00 mol of nitrogen occupies 11.2 dm^3 at 0 °C. Calculate the pressure in kPa.

1. Convert temperature: T = 0 + 273 = **273 K**
2. Convert volume: 11.2 dm^3 ÷ 1000 = **0.0112 m^3**
3. Rearrange: P = nRT ÷ V
4. Substitute: P = (1.00)(8.314)(273) ÷ 0.0112
5. P = 202 654 Pa ÷ 1000

P = **202.65 kPa**

WORKED — MOLAR MASS, MOLES AND PARTICLES

Three quantities from one sample

A sample contains 76.00 g of fluorine gas, F_2. Calculate its molar mass, the number of moles present, and the number of molecules.

1. Molar mass: F_2 has two fluorine atoms, so M = 2 × 19.00 = **38.00 g/mol**
2. Moles: n = m ÷ M = 76.00 ÷ 38.00 = **2.00 mol**
3. Molecules: N = n × 6.02 × 10^{23}
4. N = 2.00 × 6.02 × 10^{23}
5. **The question asks for molecules, not atoms.** Each F_2 holds two atoms, so the atom count would be twice this

M = 38.00 g/mol
n = 2.00 mol
N = **1.20 × 10^{24} molecules**

WORKED — IDENTIFYING A GAS FROM ITS DENSITY DATA

Which gas is it

0.71 g of an unknown gas occupies 250 cm^3 at 101 kPa and 30.7 °C. Identify the gas.

1. Convert every quantity to base units: V = 250 cm^3 = **0.00025 m^3**, P = 101 kPa = **101 000 Pa**, T = 30.7 + 273 = **303.7 K**
2. Find the moles: n = PV ÷ RT
3. n = (101 000 × 0.00025) ÷ (8.314 × 303.7) = 25.25 ÷ 2524.9
4. n = **0.0100 mol**
5. Molar mass: M = mass ÷ n = 0.71 ÷ 0.0100 = **71 g/mol**
6. Match it: Cl_2 is 2 × 35.45 = 70.90 g/mol

The gas is **chlorine, Cl_2**

RELATIVE ATOMIC MASSES IN COMMON USE

ELEMENT	A_R	ELEMENT
H hydrogen	1.01	S sulfur **32.07**
C carbon	12.01	Cl chlorine **35.45**
N nitrogen	14.01	K potassium **39.10**
O oxygen	16.00	Ca calcium **40.08**
F fluorine	19.00	Fe iron **55.85**
Na sodium	22.99	Cu copper **63.55**
Mg magnesium	24.31	Zn zinc **65.38**
Al aluminium	26.98	Br bromine **79.90**
P phosphorus	30.97	I iodine **126.90**
He helium	**4.00**	**use 4.00, not 4.003**

WORKED — PERCENTAGE CHANGE IN VOLUME

Volume change when pressure and temperature both alter

1.5 mol of gas is at 101.3 kPa and 298 K. It is taken to 54.0 kPa and 258 K. Calculate the percentage increase in volume.

1. **There is no combined gas law here**, so either apply PV = nRT twice, or take the ratio directly
2. By the ratio: $V_2/V_1 = (P_1 \div P_2) \times (T_2 \div T_1)$
3. Pressure factor: 101.3 ÷ 54.0 = 1.876
4. Temperature factor: 258 ÷ 298 = 0.866
5. V_2/V_1 = 1.876 × 0.866 = **1.624**
6. Percentage increase = (1.624 − 1) × 100

Volume increases by **62.4%**

WORKED — BALANCING, THEN READING THE MOLE RATIO

From an unbalanced equation to an amount

Balance $C_5H_{12} + O_2 \rightarrow CO_2 + H_2O$, then state how many moles of oxygen react with 0.30 mol of C_5H_{12}.

1. Balance carbon first: 5 C on the left, so **5**CO_2
2. Balance hydrogen next: 12 H on the left, so **6**H_2O
3. Count the oxygen now fixed on the right: 5(2) + 6(1) = **16** atoms
4. Oxygen arrives as O_2, so 16 ÷ 2 = **8**O_2
5. Balanced: C_5H_{12} + **8**O_2 → **5**CO_2 + **6**H_2O
6. Ratio $C_5H_{12} : O_2$ = 1 : 8, so 0.30 mol needs 0.30 × 8

2.40 mol of oxygen. Balance **before** reading any ratio

SOLUTIONS — concentration, dilution, titration, limiting reagent

SOLUTIONS, DILUTION AND TITRATION

SIGNATURE	METHOD	EXAMPLE
"molarity", mol/L, [X]	M = n ÷ V, **V in litres**	18.02 g glucose (M = 180.18) in 250 mL → 0.100/0.250 = **0.400 M**
the other three units	**% by mass** = solute ÷ **solution** **g/L** = mass ÷ volume **mol/kg** = moles ÷ **solvent** in kg	**molality divides by the solvent**; the others by the whole solution
"diluted to", a volumetric flask	$M_1V_1 = M_2V_2$	**V_2 is the flask capacity**, not the water added. 25 mL into a 250 mL flask → V_2 = 250
dilution factor, serial dilution	DF = $V_{final} \div V_{initial}$	serial dilutions multiply: 10 × 20 = **200**
titration, a burette table	1. discard any trial marked **Rough** 2. volume used = final − initial, per trial 3. average the **Accurate** trials 4. n = M × V for the known solution 5. apply the **mole ratio**, divide by the other volume	the species with the **larger coefficient** takes the larger mole count
pH given in place of a concentration	$[H^+] = 10^{-pH}$; for a monoprotic acid this is its molarity	pH 1.50 → [HCl] = **0.0316 M**

WORKED — TITRATION WITH A 2 : 1 RATIO

Concentration of NaOH by titration against H_2SO_4

10.00 mL of NaOH is titrated against 0.0252 mol/L H_2SO_4. Titres: 15.10, 15.00, 14.90, 15.00 mL, all recorded as accurate.
$2\ NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2\ H_2O$

1. Average the accurate titres: (15.10 + 15.00 + 14.90 + 15.00) ÷ 4 = **15.00 mL**
2. Moles of the known solution: n(H_2SO_4) = 0.0252 × 0.01500 L = **3.78 × 10^{-4} mol**
3. Mole ratio from the equation is NaOH : H_2SO_4 = **2 : 1**, so **multiply** by 2
4. n(NaOH) = 2 × 3.78 × 10^{-4} = 7.56 × 10^{-4} mol
5. Divide by the NaOH volume: 7.56 × 10^{-4} ÷ 0.01000 L

[NaOH] = 0.0756 mol/L

WORKED — LIMITING REAGENT AND PRODUCT MASS

Mass of copper deposited

4.20 g of zinc is added to 200 mL of 0.250 mol/L copper(II) sulfate. $Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$. Calculate the mass of copper formed.

1. Moles of zinc: n = 4.20 ÷ 65.38 = **0.0642 mol**
2. Moles of copper(II) sulfate: n = M × V = 0.250 × **0.200 L** = **0.0500 mol**
3. The ratio is **1 : 1**, so each would give that many moles of Cu
4. $CuSO_4$ gives the smaller amount, so **$CuSO_4$ is limiting** — even though there is less zinc by mass
5. n(Cu) = 0.0500 mol, and M(Cu) = 63.55 g/mol
6. m = 0.0500 × 63.55

Mass of copper = 3.18 g

LIMITING REAGENT

SIGNATURE	METHOD	EXAMPLE
two amounts given, or "which is limiting"	1. for **each** reactant, find how much product it alone would give 2. the smaller product amount identifies the limiting reagent 3. the other is in excess	4 mol CO (1:1) → 4 6 mol H_2 (2:1) → 3 H_2 gives less → **H_2 limiting**
"how much remains"	excess = starting amount − amount **consumed**	0.60 mol H_2, 0.40 consumed → **0.20 mol left**
common error	**the smaller starting amount is not necessarily limiting**	Zn 0.0642 mol vs $CuSO_4$ 0.0500 mol at 1:1 → $CuSO_4$ limits
one amount given as a solution	convert first: n = M × V	0.250 M × 0.200 L = **0.0500 mol**

TITRATION TECHNIQUE, AND STANDARDISATION

Why NaOH is standardised: it cannot be weighed accurately because it is **hygroscopic** and absorbs **CO_2** from the air, so its concentration must be found against an acid of known concentration.

Technique: read the meniscus at the **bottom**; no air bubble in the burette tip; add dropwise near the endpoint; phenolphthalein as indicator.

MODEL ANSWER — STANDARDISATION

The concentration of sodium hydroxide cannot be determined by weighing because it is **hygroscopic** and absorbs **carbon dioxide** from the air, so its mass is not reliable. It is therefore standardised against **a solution of known concentration**.

Outside the syllabus: STP and 22.4 L/mol; Gay-Lussac; the combined gas law; percent composition; empirical and molecular formula; ppm and % w/v.

WORKED — DILUTION FROM A STOCK SOLUTION

Concentration after dilution

25 mL of 6.0 mol/L glucose is pipetted into a 250 mL volumetric flask and made up to the mark. Calculate the moles transferred and the final concentration.

1. Moles pipetted: n = M × V = 6.0 × **0.025 L** = **0.15 mol**
2. **Dilution adds solvent, not solute**, so this 0.15 mol is unchanged by the dilution
3. Apply $M_1V_1 = M_2V_2$. Here V_1 = 25 mL and **V_2 = 250 mL, the flask capacity** — not the 225 mL of water added
4. M_2 = (6.0 × 25) ÷ 250
5. Check against the moles: 0.15 mol ÷ 0.250 L = 0.60 mol/L, which agrees

n = 0.15 mol transferred
final concentration = **0.60 mol/L**

MODEL WORDING — SOLUTIONS AND TECHNIQUE

STATING A TITRATION RESULT

The concentration is **0.0756 mol/L**. From the balanced equation the ratio of **NaOH** to **H_2SO_4** is **2 : 1**, so the moles of **NaOH** are **twice** those of the acid.

WHY A ROUGH TITRE IS DISCARDED

The rough titre is excluded because it is only an approximate value used to locate the endpoint. Averaging the **accurate** titres gives **15.00 mL**.

THE FOUR CONCENTRATION UNITS, SIDE BY SIDE

UNIT	DIVIDED BY	WORKED
percent by mass, %	mass of the whole **solution**	0.30 g in 0.30 + 0.50 g → 0.30/0.80 = **37.50%**
mass concentration, g/L	volume of solution in **litres**	0.35 g in 1 L → **0.35 g/L**
molality, mol/kg	**mass of the SOLVENT in kg**	0.30 mol in 0.50 kg water → **0.60 mol/kg**
molarity, mol/L	volume of **solution** in litres	0.30 mol in 0.60 L → **0.50 mol/L**
the one difference	**molality divides by the solvent; every other unit divides by the whole solution**	the same 0.30 mol gives four different numbers

WORKED — CONCENTRATION FROM A TITRATION WITH PH DATA

Using pH to obtain the titrant concentration first

A burette holds hydrochloric acid of pH 1.50. It is titrated against 25.00 mL of NaOH. Titres: 8.70, 8.80, 8.60, 8.90 mL. $HCl + NaOH \rightarrow NaCl + H_2O$. Find [NaOH].

1. Convert the pH: $[H^+] = 10^{-1.50}$ = **0.0316 mol/L**
2. HCl is monoprotic, so [HCl] = $[H^+]$ = **0.0316 mol/L**
3. Average the titres: (8.70 + 8.80 + 8.60 + 8.90) ÷ 4 = **8.75 mL**
4. Moles of acid: n = 0.0316 × 0.00875 L = **2.77 × 10^{-4} mol**
5. The ratio is **1 : 1**, so n(NaOH) is the same
6. Divide by the NaOH volume: 2.77 × 10^{-4} ÷ 0.02500 L

[NaOH] = 0.0111 mol/L

ACIDS — equilibrium, Kc, pH, weak acids, buffers

"FIND THE PH OF..." — IDENTIFYING THE CASE

WHAT IS SHOWN	CASE	ROUTE
single arrow →, a named acid, no K quoted	**strong acid**	$[H^+]$ = (number of H^+) × [acid], then pH = −log$[H^+]$
NaOH, KOH, "alkaline"	**strong base**	**two steps.** $[OH^-]$ → pOH = −log$[OH^-]$ → **pH = 14 − pOH**
double arrow ⇌, or K_a quoted	**weak acid**	the **% dissociated** is required, and is always supplied
a **% dissociated** figure	weak acid, direct	$[H^+]$ = % × [acid], then −log
two solutes, the second a **sodium salt of the first**	**buffer**	pH = pK_a + log([salt] ÷ [acid])
weak **base** with its salt	**alkaline buffer**	**yields pOH.** Then pH = 14 − pOH
strong acid with its salt (HCl + NaCl)	**not a buffer**	treat as the strong acid alone
weak acid with an **unrelated** salt	**not a buffer**	treat as the weak acid alone
the two concentrations are equal	**pH = pK_a exactly**	log(1) = 0; the log term vanishes

FIND THE pH OF ...

how many solutes?

ONE single arrow, a named acid
STRONG ACID
[H+] = n x [acid], then -log

ONE NaOH, KOH, alkaline
STRONG BASE
-log[OH-], then 14 - pOH

ONE double arrow, or Ka quoted
WEAK ACID
[H+] = % x [acid], -log

TWO the salt of that same acid
ACIDIC BUFFER
pKa + log(salt/acid)

TWO the salt of that same base
ALKALINE BUFFER
pOH first, then 14 - pOH

TWO unrelated salt, or strong acid
NOT A BUFFER
use the acid alone

Equal concentrations: pH = pKa exactly

PH CONVERSIONS AND STRONG ACIDS

SIGNATURE	METHOD	EXAMPLE
the four relationships	pH = −log$[H^+]$ $[H^+]$ = 10^{-pH} pOH = −log$[OH^-]$ $[OH^-]$ = 10^{-pOH} **pH + pOH = 14**	any quantity reaches any other in at most two steps
monoprotic strong acid: HCl, HNO_3, HBr	$[H^+]$ = **1** × [acid]	0.03 M HCl → pH = **1.52**
diprotic strong acid: H_2SO_4	$[H^+]$ = **2** × [acid]	0.015 M → $[H^+]$ = 0.030 → pH = **1.52**, identical
H_2CO_3, H_2S	**diprotic but weak**	the doubling applies to **strong** diprotic acids only
strong base	1. $[OH^-]$ = [base] 2. pOH = −log$[OH^-]$ 3. **pH = 14 − pOH**	0.04 M NaOH → pOH 1.40 → pH **12.60**
weak acid, % dissociation supplied	1. $[H^+]$ = % × initial concentration 2. pH = −log$[H^+]$ 3. K_a = $[H^+][A^-]$ ÷ [HA] if required	0.010 M at 5% → $[H^+]$ = 5.0 × 10^{-4} → pH **3.30**

PH CONVERSIONS AND STRONG ACIDS

SIGNATURE	METHOD	EXAMPLE
a weak and a strong acid at the same pH	0.010 M weak and 0.0005 M strong both give **3.30**	**pH cannot distinguish strong from weak.** Only K_a can

BUFFERS

SIGNATURE	METHOD	EXAMPLE
is the mixture a buffer	**1.** is the acid (or base) **weak**? strong → no **2.** are **both** partners present? acid alone → no **3.** is the salt the conjugate of **that same** acid?	all three conditions must hold
acidic buffer	**pH = pK_a + log([salt] ÷ [acid])**	**salt in the numerator.** pK_a = −log K_a
alkaline buffer	pOH = pK_b + log([salt] ÷ [base])	**yields pOH.** Then pH = 14 − pOH
check the direction each time	more salt than acid → ratio > 1 → log positive → pH **above** pK_a	a result moving the other way indicates an inverted ratio
a small amount of **acid** is added	the **basic** partner consumes it	$H^+ + CH_3COO^- \rightarrow CH_3COOH$
a small amount of **base** is added	the **acidic** partner consumes it	$CH_3COOH + OH^- \rightarrow H_2O + CH_3COO^-$
"buffering capacity"	higher **concentration** of both components, as near **equal** as possible	two buffers may share a pH and differ greatly in capacity
selecting an acid for a target pH	at equal concentrations pH = pK_a	target 8.6 → the acid with pK_a **8.60**
finding a required ratio	[salt] ÷ [acid] = $10^{(pH - pK_a)}$	blood: $10^{(7.40-6.1)}$ → about **20 : 1**

WORKED — BUFFER PH, AND THE EFFECT OF ADDING ACID

pH of a nitrous acid buffer

A buffer contains 0.80 mol/L HNO_2 and 0.50 mol/L $NaNO_2$. $K_a(HNO_2)$ = 4.0 × 10^{-4}. Calculate its pH, and state what happens when a little HCl is added.

1. Convert K_a: pK_a = −log(4.0 × 10^{-4}) = **3.40**
2. Identify the parts: acid = HNO_2 at 0.80 M, salt = $NaNO_2$ at 0.50 M
3. Substitute salt over acid: pH = 3.40 + log(0.50 ÷ 0.80)
4. log(0.625) = −0.204, so pH = 3.40 − 0.204
5. Check the direction: less salt than acid, so pH sits **below** pK_a — consistent

pH = 3.20
added H^+ is consumed by NO_2^- → HNO_2, so the pH is nearly unchanged

EQUILIBRIUM AND K_c

SIGNATURE	METHOD	EXAMPLE
"write the K_c expression"	**1. balance the equation first** **2.** products in the numerator, reactants in the denominator **3.** each concentration raised to its coefficient **4.** multiply, never add	$CH_4 + 2H_2S \rightleftharpoons CS_2 + 4H_2$ K_c = $[CS_2][H_2]^4$ ÷ $([CH_4][H_2S]^2)$

EQUILIBRIUM AND K_c

SIGNATURE	METHOD	EXAMPLE
"value and units"	units = $M^{\Delta n}$, where Δn = (powers above) − (powers below)	Δn = 0 → **no units**
Q_c against K_c	identical algebra	Q_c at any instant; equal to K_c only **at equilibrium**
"essentially only product or reactant"	K > 10^2 product K < 10^{-2} reactant between: comparable	**the boundary is 100.** K = 97 is comparable
add reactant, or remove product	shifts **right**	K_c unchanged
increase pressure	shifts toward **fewer moles of gas**	equal moles → no shift. K_c unchanged
increase temperature	shifts toward the side that absorbs heat	**the only change that alters K_c**
catalyst	no shift	no change to position, to K_c, or to yield
heat written into the equation	treat **heat as a species**	products **+ heat** means exothermic, so heating shifts **left**

ACIDS — SCOPE NOTES AND MODEL WORDING

MODEL ANSWER — LE CHATELIER

The equilibrium shifts to the **left**, because increasing the **temperature** favours the **endothermic** direction, which absorbs the added heat. The amount of **CH_4** therefore **decreases**.

MODEL ANSWER — BUFFER IDENTIFICATION

This mixture **is** a buffer, because it contains a weak acid, **HNO_2**, together with its conjugate base, **NO_2^-**, supplied by the salt. Added acid is removed by the conjugate base and added alkali by the weak acid, so the pH changes very little.

Outside the syllabus: ICE tables, K_p and partial pressures, K_w as a symbol, K_a × K_b = K_w, the "x is small" approximation, pH of a weak base from K_b alone, pH of a salt solution, and omitting pure solids from K_c.

Two printing errors in the notes. One equilibrium equation reads **$PCl_5 \rightleftharpoons PCl_5 + Cl_2$**, which is impossible; the left side should be PCl_3. Another gives **$SO_2 + O_2 \rightleftharpoons 2SO_3$** unbalanced; it should read $2SO_2$.

pK_a of ethanoic acid appears once as 4.76 and elsewhere as **4.74**. Use **4.74**.

WORKED — A WEAK ACID AND A STRONG ACID AT THE SAME PH

pH, K_a, and what pH cannot tell you

A 0.010 mol/L solution of the weak acid HY is 5% dissociated. Calculate its pH and K_a. A 0.0005 mol/L solution of the strong acid HX is also measured. Comment on what pH reveals about acid strength.

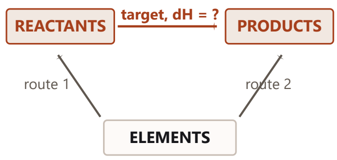
1. Dissociated fraction: $[H^+]$ = $[Y^-]$ = 5% × 0.010 = **5.0 × 10^{-4} mol/L**
2. Undissociated acid remaining: [HY] = 95% × 0.010 = **0.0095 mol/L**
3. pH = −log(5.0 × 10^{-4}) = **3.30**
4. K_a = $[H^+][Y^-]$ ÷ [HY] = $(5.0 \times 10^{-4})^2$ ÷ 0.0095 = **2.6 × 10^{-5}**
5. For the strong acid, dissociation is complete: $[H^+]$ = 0.0005, so pH = −log(0.0005) = **3.30**
6. **Both solutions have pH 3.30**, yet one acid is weak and the other strong

pH depends on concentration as well as strength, so it **cannot** distinguish a weak acid from a strong one. Only **K_a** can

ENERGY — enthalpy, Hess, entropy, free energy, spontaneity

HESS'S LAW — COMBINING EQUATIONS

SIGNATURE	METHOD	EXAMPLE
"calculate $\Delta_{rxn}H$", then **"given that"** and two or three equations	**1.** write the **target** equation **2.** for each species in it, ask whether it is on the correct side **3.** adjust each supplied equation by one of the three moves below **4.** add them and **cancel** what appears on both sides **5.** if the result is not the target, a move is wrong	the adjusted ΔH values then add up
species already correct	**keep** the equation	ΔH unchanged
species on **the wrong side**	**reverse** the equation	**flip the sign of ΔH.** Magnitude unchanged
coefficient wrong by a factor	**multiply** or halve the equation	**multiply ΔH by the same factor.** Sign unchanged
both faults at once	reverse **and** scale, independently	reverse gives +966, then halving gives **+483**



Both routes begin and end together, so both cost the same. Going against an **arrow flips that dH; scaling scales it.**

WORKED — HESS'S LAW WITH TWO REVERSALS

Enthalpy of formation of ammonium chloride

Calculate $\Delta_{rxn}H$ for $NH_3(g) + HCl(g) \rightarrow NH_4Cl(s)$. Given: (1) $\frac{1}{2}N_2 + 1\frac{1}{2}H_2 \rightarrow NH_3$, ΔH = −46.1 kJ (2) $\frac{1}{2}H_2 + \frac{1}{2}Cl_2 \rightarrow HCl$, ΔH = −92.3 kJ (3) $\frac{1}{2}N_2 + 2H_2 + \frac{1}{2}Cl_2 \rightarrow NH_4Cl$, ΔH = −314.4 kJ

1. NH_3 must be a **reactant**, but is a product in (1). **Reverse (1)** → ΔH = **+46.1**
2. HCl must be a **reactant**, but is a product in (2). **Reverse (2)** → ΔH = **+92.3**
3. NH_4Cl is already a product in (3). **Keep (3)** → ΔH = **−314.4**
4. No coefficient is wrong, so nothing is scaled
5. Add: the $\frac{1}{2}N_2$, $2H_2$ and $\frac{1}{2}Cl_2$ appear on both sides and cancel, leaving $NH_3 + HCl \rightarrow NH_4Cl$
6. Sum the enthalpies: 46.1 + 92.3 − 314.4

$\Delta_{rxn}H$ = **−176.00 kJ**, exothermic

ENTHALPY FROM TABLES

SIGNATURE	METHOD	EXAMPLE
a table of **Δ_fH** values	**$\Delta_{rxn}H$ = Σn(products) − Σn(reactants)**	n is the coefficient from the balanced equation
an **element** appears	its Δ_fH is **zero**	including the diatomics H_2 N_2 O_2 F_2 Cl_2 Br_2 I_2
worked	$4FeS_2 + 11O_2 \rightarrow 2Fe_2O_3 + 8SO_2$	[2(−824) + 8(−297)] − [4(−178) + 0] = −4024 + 712 = **−3312.00 kJ**

ENTHALPY FROM TABLES

SIGNATURE	METHOD	EXAMPLE
a built-in check	the same reaction by the Hess route and the formation route	both give **−176.00 kJ**; disagreement indicates an error
a table of **bond enthalpies**	**ΔH = breaking (+) + forming (−)**	**reactants first**, the reverse of the formation route
worked, combustion of propanone	break 6(C–H) + 2(C–C) + 1(C=O) + 4(O=O) = +5957 form 6(C=O) + 6(O–H) = −7608	**−1651.00 kJ**. A positive combustion result is always wrong
bond counting	each CO_2 holds **two** C=O, each H_2O **two** O–H, each O_2 one O=O to break	3 CO_2 therefore gives 6, not 3

ENTROPY

SIGNATURE	METHOD	EXAMPLE
"increase or decrease in entropy", no numbers	**1.** count the **particles**: more on the right raises ΔS **2.** compare **states**: solid < liquid < aqueous < gas **3.** state the sign	**both tests are required** – one examined case has no particle change and turns on the state test alone
rapid check	count moles of **gas**	gas up → positive, gas down → negative
worked, the case that turns on states	NaOH(s) + acid(aq) → salt(aq) + H_2O(l). Particles 2 → 2	**solid → aqueous**, so ΔS is **positive**
a table of **S°** values	**ΔS = ΣS(products) − ΣS(reactants)**	**elements are not zero here.** O_2 has S° = 205.0
worked	$N_2 + 3H_2 \rightarrow 2NH_3$	2(193.0) − [191.5 + 3(131.0)] = **−198.50 J mol⁻¹K⁻¹**

FREE ENERGY AND SPONTANEITY

SIGNATURE	METHOD	EXAMPLE
ΔH and ΔS both given, and the word **spontaneous**	**1.** convert T to **kelvin**: K = °C + 273 **2. check the ΔS unit**; if J, divide by 1000 **3.** ΔG = ΔH − TΔS **4.** state the sign and the verdict	ΔG < 0 spontaneous > 0 not spontaneous = 0 equilibrium
the unit trap	one examined question gives ΔH in **kJ** with ΔS in **J**; another gives ΔS already in **kJ**	**read the unit every time**
the double negative	−T × (−ΔS) is **plus**	−92.2 + 59.30 = −32.90, not −151.5
worked, ice melting	ΔH = +6.01 kJ, ΔS = 22.0 J = **0.0220 kJ**, T = 298 K	6.01 − (298)(0.0220) = **−0.55** kJ/mol, spontaneous
"at what temperature"	set ΔG = 0, so **T = ΔH ÷ ΔS**	6.01 ÷ 0.0220 = **273.2 K**, the melting point of ice
which side of the crossover	both signs + → spontaneous **above** T both − → spontaneous **below** T	the mixed cases never change verdict

WORKED — SPONTANEITY AT TWO TEMPERATURES

The Haber process

For $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$, ΔH = −92.2 kJ/mol and ΔS = −0.199 kJ mol⁻¹K⁻¹. Determine whether the reaction is spontaneous at 298 K and at 773 K.

1. ΔS is already in **kJ**, so no conversion is needed
2. At 298 K: ΔG = −92.2 − (298)(−0.199)
3. **Two negatives give a plus:** −92.2 **+** 59.30 = **−32.90 kJ/mol**, spontaneous
4. At 773 K: ΔG = −92.2 − (773)(−0.199) = −92.2 + 153.83
5. = **+61.63 kJ/mol**, not spontaneous
6. Crossover: T = ΔH ÷ ΔS = −92.2 ÷ −0.199 = **463.3 K**, which explains both results

Spontaneous at 298 K, not at 773 K. Both ΔH and ΔS negative → spontaneous only **below** 463.3 K

SPONTANEITY — THE FOUR CASES

	ΔH NEGATIVE (EXOTHERMIC)	ΔH POSITIVE (ENDOTHERMIC)
ΔS positive	**spontaneous at all temperatures**	spontaneous at **high** T, once TΔS is large enough
ΔS negative	spontaneous at **low** T, while TΔS is small	**never spontaneous**
reason	only the **TΔS** term varies with temperature	where ΔH and ΔS agree in sign, temperature cannot change the verdict

ENERGY — UNITS, CONVENTIONS AND MODEL WORDING

MODEL ANSWER — SPONTANEITY

The reaction is **spontaneous** at **298 K**, because ΔG = ΔH − TΔS = **−32.90 kJ/mol**, which is **negative**. A negative free energy change indicates a spontaneous process.

MODEL ANSWER — ENTROPY CHANGE

Entropy **increases**, because the number of particles rises from **2** to **3**, and because **a gas** is produced where there was none. A gas has the greatest number of possible arrangements.

Units: tabulated data are in **kJ/mol**; calculated answers in **kJ**, unless an enthalpy **of formation** is asked for, which is per mole. Entropy tables are in **J**; divide by 1000 before combining with an enthalpy.

Signs: −ΔH exothermic, heat released. +ΔH endothermic, heat absorbed. Breaking bonds absorbs (+), forming bonds releases (−).

"Gives out 81.67 kJ" means ΔH = −81.67. The minus sign is not written in the question.

Thermodynamics answers are given to **2 decimal places**. **Calorimetry (q = mcΔT) is outside the syllabus.**

WORKED — ENTHALPY FROM BOND ENTHALPIES

Combustion of propanone

$CH_3COCH_3(g) + 4O_2(g) \rightarrow 3CO_2(g) + 3H_2O(g)$. Bond enthalpies in kJ/mol: C–H 412, C–C 348, C=O 805, O=O 496, O–H 463.

1. Count the bonds broken. Propanone is $H_3C–CO–CH_3$: **6** C–H, **2** C–C, **1** C=O. Plus **4** O=O in the oxygen
2. Breaking = 6(412) + 2(348) + 805 + 4(496) = 2472 + 696 + 805 + 1984 = **+5957 kJ**
3. Count the bonds formed. **Each CO_2 holds two C=O**, so 3 CO_2 gives **6** C=O. Each H_2O holds two O–H, so 3 H_2O gives **6** O–H
4. Forming = 6(805) + 6(463) = 4830 + 2778 = **−7608 kJ**
5. ΔH = breaking + forming = 5957 + (−7608)

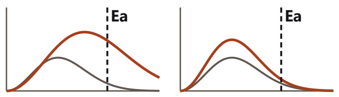
$\Delta_{rxn}H$ = **−1651.00 kJ**. A combustion returning a positive value has the two terms the wrong way round

RATES — collisions, activation energy, rate laws, catalysis

KINETICS

SIGNATURE	METHOD	EXAMPLE
"why does X speed it up"	every reason ends the same way: **more successful collisions**	surface area, concentration, pressure → more **frequent** collisions
temperature	the only factor that does **both**	more frequent collisions **and** a bigger proportion above E_a
a **catalyst**	an **alternative route with lower E_a**	more molecules clear the lower bar. It is not used up, and **k increases**
a Maxwell-Boltzmann figure	molecules that can react = the **area under the curve to the RIGHT of E_a**	shade it, then compare the shaded areas
what moves on that figure	**temperature** shifts the curve right **concentration** makes it taller, same place **catalyst** moves the E_a line, not the curve	**three different changes. Do not blur them**
"rate = $k[A]^m[B]^n$"	the orders come from **experiment**, never from the coefficients	**one exception:** in an **elementary step** of a mechanism, order = coefficient
find the order from a rates table	find two runs where **only one** concentration changed. Rate unchanged → zero order doubles → first quadruples → second	
find the order from **half-life**	t½ **constant** → first order t½ halves each time → zero t½ doubles each time → second	constant 12 s each halving → **first order**
units of k in 3 seconds	$M^{(1 - \text{overall order})}\ s^{-1}$	overall 1 → s^{-1} overall 3 → $M^{-2}\ s^{-1}$
Arrhenius, two conditions	$k = Ae^{-Ea/RT}$, so with A constant $\ln(k_2/k_1) = E_{a1}/RT_1 - E_{a2}/RT_2$	**°C → K first.** 400°C is **673 K**. R is given in **kJ**, so E_a in kJ goes straight in

higher TEMPERATURE **higher CONC.**

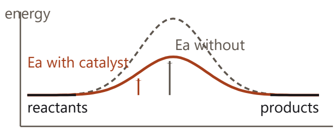


curve SHIFTS right
more molecules past Ea

curve rises, same spot
more molecules past Ea

Able to react = the area under the curve to the RIGHT of Ea.

A catalyst moves the Ea line left;
the curve itself does not move.



The catalyst lowers only the hump.

Reactant and product levels do not move,
so the enthalpy change is unchanged.

MODEL WORDING — REDOX, CELLS AND RATE

IDENTIFYING AN AGENT

The oxidising agent is **MnO_4^-**, because the oxidation state of **manganese** falls from **+7** to **+2**. It gains electrons and is itself reduced, so it oxidises the other species.

ELECTRON DIRECTION

Electrons flow from **chromium** to **nickel**, because **chromium** is the anode, where oxidation releases electrons. They travel through the external circuit to the cathode.

RATE EXPLANATION

The rate increases, because raising the **temperature** increases the proportion of molecules with energy greater than the activation energy, and also increases the frequency of collisions. This increases the number of **successful collisions** per second.

AGREE OR DISAGREE

Yes. The pump moves **two different** species in **opposite** directions, which is the definition of an antiporter.

WORKED — ORDER OF REACTION FROM HALF-LIFE

Finding the order, then testing a mechanism

The concentration of G falls 0.40, 0.20, 0.10, 0.05, 0.025, 0.0125 mol/L at 0, 12, 24, 36, 48, 60 s. Determine the order with respect to G, and decide which mechanism is consistent: (I) slow step G + H_2 → GH + H, or (II) slow step 2G → G_2.

1. Read the half-lives off the table. 0.40 → 0.20 takes **12 s**
2. 0.20 → 0.10 takes **12 s**; 0.10 → 0.05 takes **12 s**
3. The half-life is **constant** and does not depend on the starting concentration
4. A constant half-life is the signature of **first order**
5. **In an elementary step only**, the order equals the coefficient. Mechanism I gives rate = $k[G][H_2]$, first order in G
6. Mechanism II gives rate = $k[G]^2$, second order in G, which contradicts the data

First order in G, and mechanism I is the consistent one

WORKED — RATE LAW, AND THE EFFECT OF CHANGING A CONCENTRATION

Rate constant and its units

For a reaction, rate = $k[P][Q]^2$. When [P] = 0.100 mol/L and [Q] = 0.400 mol/L the rate is 0.12 mol $L^{-1}\ s^{-1}$. State the orders, calculate k with its units, and give the effect of doubling each concentration.

1. Orders read straight off the exponents: **first** in P, **second** in Q, **third** overall
2. Rearrange: k = rate ÷ $([P][Q]^2)$
3. k = 0.12 ÷ (0.100×0.400^2) = 0.12 ÷ 0.0160
4. Units: k carries $M^{(1 - \text{overall order})}\ s^{-1}$, so for third order that is **$M^{-2}\ s^{-1}$**
5. Doubling [P]: first order, so the rate changes by 2^1 = **×2**
6. Doubling [Q]: second order, so 2^2 = **×4**

k = 7.5 $M^{-2}\ s^{-1}$
doubling P doubles the rate, doubling Q quadruples it

WORKED — ARRHENIUS WITH A CATALYST

Rate constant at a second temperature

At 400 °C with E_a = 30 kJ/mol, k = $3.7 \times 10^{-5}\ s^{-1}$. Find k at 450 °C where a catalyst lowers E_a to 20 kJ/mol. R = 8.314×10^{-3} kJ $mol^{-1}\ K^{-1}$, and A is unchanged.

1. **Convert to kelvin first:** 400 °C = **673 K**, 450 °C = **723 K**
2. R is already in **kJ**, so E_a in kJ/mol goes in without conversion
3. With A constant, $\ln(k_2/k_1) = E_{a1}/RT_1 - E_{a2}/RT_2$
4. First term: 30 ÷ $(8.314 \times 10^{-3} \times 673)$ = 5.362
5. Second term: 20 ÷ $(8.314 \times 10^{-3} \times 723)$ = 3.327
6. $\ln(k_2/k_1)$ = 5.362 − 3.327 = 2.034, so $k_2/k_1 = e^{2.034}$ = 7.65
7. $k_2 = 3.7 \times 10^{-5} \times 7.65$

k_2 = 2.83 × $10^{-4}\ s^{-1}$

WHAT CHANGES THE RATE, AND WHY

FACTOR	MECHANISM	WHICH OF THE TWO
surface area	smaller particles expose more surface, so collisions are more frequent	more collisions
concentration	more molecules per unit volume, so collisions are more frequent	more collisions
pressure (gases)	the same effect as raising concentration	more collisions
temperature	a greater proportion exceed E_a, **and** collisions become more frequent	**both**
catalyst	an alternative route of lower E_a, so more molecules clear the barrier	lowers the bar
the sentence that scores	increase in X → increase in [frequency of collisions / proportion above E_a] → more **successful collisions** → rate increases	reusable for any factor

WORKED — ORDER FROM A TABLE OF INITIAL RATES

Reading two orders off experimental data

For A + B → products, doubling [A] at fixed [B] doubles the rate. Doubling [B] at fixed [A] leaves the rate unchanged. Write the rate law and give the overall order.

1. Compare the two runs where **only [A]** changed: concentration ×2, rate ×2
2. Rate × 2^1 when concentration × 2 means **first order** in A
3. Compare the two runs where **only [B]** changed: concentration ×2, rate ×1
4. Rate unchanged means **zero order** in B, so $[B]^0$ = 1 and B does not appear
5. **The orders are not the coefficients** from the balanced equation; they come only from data
6. Overall order = 1 + 0

Rate = **k[A]**, first order overall

ELECTRONS — oxidation states, redox, cells, electrolysis

OXIDATION STATES — THE RULES, IN THE ORDER THEY WIN

ASK, IN THIS ORDER	ASSIGN	EXAMPLE
is it an **uncombined element**?	**0**, and stop	S_8, O_2, P_4, Na → all **0**
is the whole species **one atom with a charge**?	= the charge, and stop	Fe^{3+} → +3
is it **Group 1, Group 2, or F**?	+1, +2, −1 — these never bend	$Mg(OH)_2$ → Mg = **+2** immediately
is it **H**?	+1, unless bonded to a **metal** → −1	NaH, CaH_2 → H = −1
is it **O**?	−2, unless **peroxide** → −1	H_2O_2 and O_2^{2-} → O = −1
is it **Cl, Br or I**?	−1 in a **binary** compound. **If O or F is also there, do NOT assign it** — leave it as the unknown	$HClO_4$ → Cl is the unknown, not −1
whatever is **left**	solve: **Σ(count × state) = the total charge**	neutral → 0 polyatomic ion → **its charge**
worked	MnO_4^- Mn + 4(−2) = **−1**	**Mn = +7**
worked	$HClO_4$ (+1) + Cl + 4(−2) = 0	**Cl = +7**
worked, the neutral-ligand trick	$Mn(H_2O)_6^{2+}$ — each H_2O is **neutral**, so contributes 0	**Mn = +2**
a fast self-check	**a state above +7 means one forgot the ion's charge**	summing MnO_4^- to 0 gives +8, which is impossible

REDOX — WHO DID WHAT TO WHOM

SIGNATURE	METHOD	EXAMPLE
"what is oxidised / reduced"	**1.** assign oxidation states on **both** sides **2.** find the atoms whose number **changed** **3.** went **up** → oxidised went **down** → reduced	nothing changed → **not a redox reaction**
"which is the oxidising agent"	**the oxidising agent is the thing that gets REDUCED**	the name says what it does to the **other** species
self-check on the sentence	the two words must be **opposites**	oxidised → **reducing** agent reduced → **oxidising** agent. If they match, it is wrong
balance a redox equation, **acid**	**1.** split into two half-equations **2.** balance everything except O and H **3.** add $\mathbf{H_2O}$ for O, then $\mathbf{H^+}$ for H **4.** add **electrons** to balance the charge **5.** scale so the electrons cancel, then add	check: **same atoms** and **same total charge** each side
balance in **base**	do all of the above **via H^+ first**, then add the same number of **OH^- to BOTH sides** and combine into H_2O	**do not start from OH^-** — it is much harder

REDOX — WHO DID WHAT TO WHOM

SIGNATURE	METHOD	EXAMPLE
"will this metal displace that one"	a metal reduces the compounds of anything **below** it in the activity series	

WORKED — BALANCING A REDOX EQUATION IN ACID

Dichromate and iodide

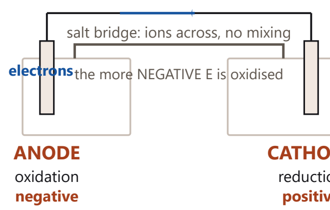
Balance $Cr_2O_7^{2-}$ + I^- → Cr^{3+} + I_2 in acidic solution.

1. Split into half-equations. Iodine: I^- → I_2. Chromium: $Cr_2O_7^{2-}$ → Cr^{3+}
2. Balance atoms other than O and H: **2**I^- → I_2 and $Cr_2O_7^{2-}$ → **2**Cr^{3+}
3. Add H_2O for oxygen, then H^+ for hydrogen: $Cr_2O_7^{2-}$ + **14**H^+ → 2Cr^{3+} + **7**H_2O
4. Add electrons to balance charge. Left is (−2 + 14) = +12, right is +6, so add **6e⁻** to the left. Oxidation: 2I^- → I_2 + **2e⁻**
5. Scale so the electrons cancel: multiply the iodide half by **3** → 6I^- → 3I_2 + 6e⁻
6. Add and cancel the electrons. Check charge: left (−2 + 14 − 6) = **+6**, right 2(+3) = **+6**

$$Cr_2O_7^{2-} + 14H^+ + 6I^- \rightarrow 2Cr^{3+} + 3I_2 + 7H_2O$$

CELLS — GALVANIC AND ELECTROLYTIC

	GALVANIC (VOLTAIC)	ELECTROLYTIC
how to tell in 3 seconds	no battery — just a voltmeter or bulb	**there is a power source** in the diagram
the reaction	**spontaneous**	non-spontaneous, must be driven
E°$_{cell}$	always **positive**	negative
anode	oxidation, **negative**	oxidation, **positive**
cathode	reduction, **positive**	reduction, **negative**
what never changes	**anode = oxidation, cathode = reduction, in BOTH**	**electrons flow anode → cathode, in BOTH.** Only the **sign** swaps
the salt bridge	a tube of electrolyte in a gel joining the two half-cells	lets **ions** flow, stops the solutions **mixing**
plating or refining	the object one want **coated or pure** is the **cathode**	metal always deposits at the cathode



Electrons always run anode to cathode.
In an ELECTROLYTIC cell the reactions are the same but the two signs swap over.

WORKED — CELL POTENTIAL AND ELECTRON DIRECTION

A chromium and nickel voltaic cell

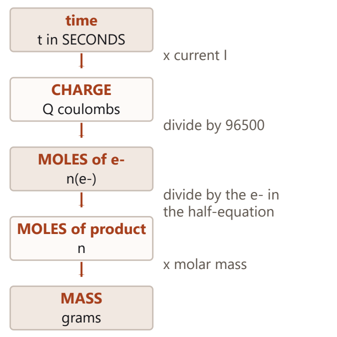
Two half-cells are joined by a salt bridge. Ni^{2+} + 2e⁻ → Ni, E° = −0.28 V. Cr^{3+} + 3e⁻ → Cr, E° = −0.74 V. Chromium is on the left. Give the electrode reactions, the overall equation, the cell potential and the direction of electron flow.

1. Compare the potentials: −0.28 is **more positive** than −0.74, so **Ni^{2+} is reduced** and is the **cathode**
2. Chromium is therefore oxidised and is the **anode**. Reverse its equation and flip the sign: Cr → Cr^{3+} + 3e⁻, E°$_{ox}$ = **+0.74 V**
3. Balance the electrons: the Cr half × **2** and the Ni half × **3** gives 6 electrons each way
4. **The E° values are not scaled.** They stay at +0.74 and −0.28
5. E°$_{cell}$ = E°$_{ox}$ + E°$_{red}$ = 0.74 + (−0.28)
6. Electrons leave the anode, which is the chromium on the **left**, and travel through the external circuit to the cathode on the right

$$2Cr + 3Ni^{2+} \rightarrow 2Cr^{3+} + 3Ni$$
E°$_{cell}$ = **+0.46 V**
electrons flow **left to right**

E° AND ELECTROLYSIS

SIGNATURE	METHOD	EXAMPLE
which one is reduced, from two E° values	the **more positive** E° is reduced → it is the **cathode**	"less positive" counts as more negative. +0.34 vs +0.80 → the +0.34 is oxidised
"calculate E°$_{cell}$"	**A105 writes this as E°$_{cell}$ = E°$_{oxidation}$ + E°$_{reduction}$**	reverse the oxidised half and **flip the sign** of its E°. The worksheet's answer table is printed in this form
is it spontaneous?	E°$_{cell}$ **positive** → yes	negative → no
a half-equation has been scaled	**E° is NOT multiplied.** Volts are per electron	scale the equation, never the volts. ×2 leaves +0.80 as **+0.80**
worked	Cr (−0.74) with Ni (−0.28)	Cr oxidised: E°$_{ox}$ = **+0.74** E°$_{cell}$ = 0.74 − 0.28 = **+0.46 V**
electrolysis: mass from current and time	**1. Q = I × t**, t in **seconds** **2.** moles of electrons = **Q ÷ 96500** **3.** divide by the electrons in the half-equation **4.** multiply by the molar mass	2 A for 1 h → Q = 7200 → 0.0746 mol e⁻ Na^+ + e⁻ → Na is 1:1 → **1.72 g**
run it backwards for "how long"	moles → ×96500 → Q → ÷ I → seconds → ÷3600 for hours	2.5 mol e⁻ at 15 A → **4.47 hours**
if the question prints an E° value	**use the question's value, not the table's**	the tables disagree on Ni (−0.25 vs −0.28)



Reverse it for "how long" or "what I".

ELECTRONS — SCOPE NOTES AND ERRATA

Use the attached potential table, not the short one printed on the slides. They come from different sources and disagree — the attached table is the one the answers follow.

Not stated in the lecture material: that a catalyst leaves ΔH, the products and the equilibrium position **unchanged**; that the salt bridge **maintains electrical neutrality and completes the circuit** (the notes says only that it lets ions through and stops mixing); what happens when E°$_{cell}$ is exactly **zero**; and cell notation such as Zn|Zn^{2+}||Cu^{2+}|Cu, which A105 does not use.

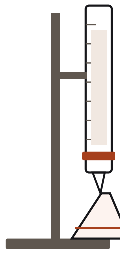
Have a periodic table to hand — one electrolysis question needs the molar masses of Hg and Cl and supplies neither.

Known typos: the activity series prints Mg twice where the second should be **Mn**; a car-battery equation prints $PbSO_2$ for $\mathbf{PbSO_4}$; one hydrogen product is labelled (s) instead of **(g)**; and the iron potential is shown as 0.44 V when it is **−0.44 V**.

LAB 1 — titration, standard solutions, serial dilution

STANDARDISING SODIUM HYDROXIDE

STAGE	DETAIL	WHY IT IS DONE THIS WAY
Establishing	the true concentration of a NaOH solution that cannot be found by weighing	NaOH is **hygroscopic** and absorbs CO_2, so the mass weighed out is not what mass of NaOH present
Method	**1.** fill the burette with the **0.010 M HCl** of known concentration **2.** pipette a fixed volume of the NaOH into a conical flask **3.** add two drops of **phenolphthalein** **4.** titrate until the pink just disappears, swirling throughout **5.** repeat until titres agree	the known solution goes in the burette because it is the one being measured out against the unknown
Assumes	the HCl really is 0.010 M, the pipette delivers its stated volume, and the indicator changes at the equivalence point	a systematic error in any of these shifts every titre the same way, so repeats will not reveal it
Common trips	**an air bubble under the tap** leaves during the run and adds to the titre reading the **top** of the meniscus including the **rough** titre in the average rinsing the conical flask with the NaOH	each of these biases the result in one direction, which is what makes them systematic rather than random
Happy path	three titres within 0.10 mL of one another, a sharp colour change on one drop	average the accurate titres, then n = MV and the 1 : 1 ratio
Sad path	**titres drift upward run to run** → a bubble left the tip, or the burette leaks **colour fades back** → CO_2 absorbed, or the end point was overshot **titres scatter randomly** → inconsistent swirling or end-point judgement	a **drift** is systematic and must be fixed; **scatter** is random and is reduced by repeating



BURETTE
the solution of KNOWN concentration goes here

read at the BOTTOM of the meniscus, to 0.05 mL

TAP
no air bubble below it, or the titre reads high

CONICAL FLASK
the UNKNOWN, pipetted, plus 2 drops of indicator

Swirl constantly, and add dropwise near the end point.

Phenolphthalein is colourless in acid and pink in alkali.

FINDING THE CONCENTRATION OF SULFURIC ACID

STAGE	DETAIL	WHY IT IS DONE THIS WAY
Establishing	the molarity of an unknown H_2SO_4 solution	it follows the first experiment, and uses the NaOH that was standardised there
Method	as before, but the balanced equation is now $\mathbf{2\ NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2\ H_2O}$	the apparatus does not change; only the arithmetic does
The one difference that matters	**the ratio is 2 : 1, not 1 : 1**	$n(NaOH) = 2 \times n(H_2SO_4)$. The species with the larger coefficient takes the larger mole count
Common trips	carrying the 1 : 1 habit over from the first experiment using the nominal NaOH concentration instead of the **standardised** one	the whole point of the first experiment was to replace the nominal value
Happy path	a result consistent with the acid label, concordant titres	quote the unit, and match the precision of the data
Sad path	**the answer is out by a factor of two** → the mole ratio was applied upside down or omitted	a factor of exactly 2 is almost always the ratio, not the arithmetic

PREPARING A STANDARD SOLUTION, AND READING THE GLASSWARE

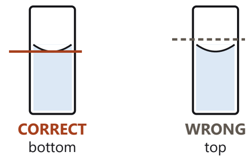
POINT	DETAIL	CONSEQUENCE IF IGNORED
choosing apparatus	**volumetric** glassware for anything that must be precise: bulb pipette, volumetric flask, burette **ordinary** for everything else: beaker, conical flask, measuring cylinder	measuring a precise volume in a beaker discards the precision the calculation assumes
the transfer	dissolve in a small volume first, transfer through a funnel, then **rinse three times** into the flask	solute left in the beaker makes the solution **too dilute**, and the error is invisible afterwards

PREPARING A STANDARD SOLUTION, AND READING THE GLASSWARE

POINT	DETAIL	CONSEQUENCE IF IGNORED
topping up	dropper for the last drops, then stopper and invert to mix	overshooting the line cannot be corrected by removing liquid
reading	bottom of the meniscus, at eye level	reading from above gives a consistently low volume

1. dissolve the solid in a small volume, in a beaker
2. transfer to the flask through a filter funnel
3. rinse the beaker and transfer, three times
4. top up to the line, dropper at the end
5. stopper and invert to mix thoroughly

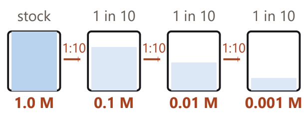
Rinsing matters: solute left behind makes the solution too dilute.



Read the BOTTOM of the meniscus, eye level.

SERIAL DILUTION

STAGE	DETAIL	NOTE
Establishing	a solution far more dilute than one step could produce	used in chemical analysis, microbiology and pharmacology
Method	transfer one part into nine parts of solvent, and repeat	each step is a factor of ten
The arithmetic	dilution factors **multiply**	10 × 10 × 10 = **1000**, in three manageable steps
Common trips	adding solvent **to** the mark rather than making up **to** the mark failing to mix between steps	an unmixed step carries its error into every step after it
Sad path	**the final concentration is out by a power of ten**	one step was skipped, or a factor was added instead of multiplied



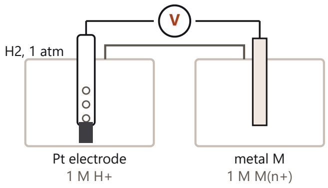
Take 1 part into 9 parts of solvent, four times over. Dilution factors MULTIPLY: 10 x 10 x 10 = 1000.

One 1000-fold step in a single flask cannot be measured accurately, which is why the chain is used.

LAB 2 — electrode potentials, electrolysis, rate methods

MEASURING A STANDARD ELECTRODE POTENTIAL

STAGE	DETAIL	WHY IT IS DONE THIS WAY
Establishing	the standard reduction potential of a metal	a potential can only be measured as a **difference**, so a reference is needed
Method	**1.** build the metal half-cell: the metal in 1 M solution of its ion **2.** connect it to a **standard hydrogen electrode** **3.** join the two with a salt bridge **4.** read the voltmeter	the hydrogen electrode is **defined** as 0.00 V, so the reading is the metal potential itself
Standard conditions	1 M solutions, 1 atm gas pressure, 298 K	E° values are only comparable at the same conditions
Common trips	omitting the salt bridge, so no circuit forms reading the magnitude and dropping the **sign**	the sign is what decides which electrode is which
Happy path	a stable reading whose sign tells you the metal is oxidised (negative) or reduced (positive) relative to hydrogen	
Sad path	**the needle does not move** → broken circuit, usually the salt bridge **the reading drifts** → concentrations are not 1 M, or the cell is running down	a drifting cell is no longer at standard conditions, so the value is not E°



The hydrogen electrode is defined as exactly 0.00 V, so the voltmeter reads the metal potential directly.

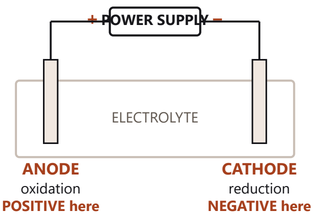
Standard conditions throughout: 1 M solutions, 1 atm gas, 298 K.

ELECTROLYSIS, PLATING AND REFINING

STAGE	DETAIL	WHY IT IS DONE THIS WAY
Establishing	a non-spontaneous change, driven by a power supply	E°_{cell} is negative, so it will not proceed alone

ELECTROLYSIS, PLATING AND REFINING

STAGE	DETAIL	WHY IT IS DONE THIS WAY
Method	two electrodes in the electrolyte, wired to a supply, for a measured current and time	the mass deposited is set by the charge passed
The chain	Q = I × t → ÷ 96500 → moles of electrons → ÷ the electrons in the half-equation → × molar mass	t in **seconds**
Plating or refining	the object being coated or purified is the **cathode**	metal always deposits at the cathode
Common trips	time left in minutes forgetting to divide by the electrons per ion, so Ag and Au are treated alike assuming the galvanic signs still apply	Ag^+ needs 1 electron, Au^{3+} needs 3
Happy path	the electrode gains mass; weighing before and after gives the amount deposited	the silver experiment in the notes works exactly this way
Sad path	**less mass than predicted** → a side reaction competed, or the current was not steady **gas at an electrode** → water was discharged in place of the intended ion	competing electrode reactions are decided by which E° is more favourable



Metal deposits at the CATHODE, so the object **being plated or purified is wired there.**

The signs reverse from a galvanic cell; the electrode reactions do not.

DETERMINING A REACTION ORDER

STAGE	DETAIL	WHY IT IS DONE THIS WAY
Establishing	the order with respect to each reactant	**orders cannot be read off the balanced equation** and must come from experiment

DETERMINING A REACTION ORDER

STAGE	DETAIL	WHY IT IS DONE THIS WAY
Method A, initial rates	run the reaction several times, changing **one** concentration at a time, and measure the initial rate each time	holding everything else fixed is what isolates the order
Reading it	concentration ×2 and rate unchanged → order **0** rate ×2 → order **1** rate ×4 → order **2**	
Method B, half-life	follow one run and measure successive half-lives	$t_{½}$ **constant** → first order $t_{½}$ halving → zero $t_{½}$ doubling → second
Common trips	comparing two runs in which **two** concentrations changed using the coefficients as the orders	the only place a coefficient gives an order is an **elementary step**
Sad path	**the orders do not reproduce the observed rate** → the wrong pair of runs was compared	a proposed mechanism is rejected when its slow step predicts the wrong order

run	[A]	[B]	rate
1	0.10	0.10	2.0
2	0.20	0.10	4.0
3	0.10	0.20	2.0

Runs 1 and 2: only [A] changed, and the rate doubled, so the order in A is 1.

Runs 1 and 3: only [B] changed, and the rate did not move, so the order in B is 0.

Compare only pairs that differ in ONE concentration.

READING ANY PRACTICAL QUESTION

MODEL ANSWER, AN ERROR QUESTION

The result would be **too high**, because **the air bubble** leaves the burette tip during the titration and is recorded as though it were **titrant delivered**. This is a **systematic** error, so repeating the titration would not remove it.

MODEL ANSWER, WHY A STEP EXISTS

The two solutions must be **standardised** first because **sodium hydroxide** absorbs moisture and carbon dioxide from the air, so its concentration cannot be calculated from the **mass weighed out**.

Systematic or random: an error that shifts every result the **same way** is systematic, and repeating will not remove it. An error that scatters results is random, and averaging reduces it. Nearly every practical question turns on which of the two is being described.